

Program 10

STEP 1 : Start MiniKube with resources

```
(1RV24MC114_VENKATESHAHD@kali)~[~/mca/devops/lab/lab7]
$ minikube start --driver=docker
minikube v1.37.0 on Debian kali-rolling
✨ Using the docker driver based on existing profile
👉 Starting "minikube" primary control-plane node in "minikube" cluster
🚢 Pulling base image v0.0.48 ...
🔄 Restarting existing docker container for "minikube" ...
🔧 Preparing Kubernetes v1.34.0 on Docker 28.4.0 ...
🔍 Verifying Kubernetes components ...
  ▪ Using image gcr.io/k8s-minikube/storage-provisioner:v5
  ▪ Using image docker.io/kubernetesui/dashboard:v2.7.0
  ▪ Using image docker.io/kubernetesui/metrics-scraper:v1.0.8
💡 Some dashboard features require the metrics-server addon. To enable all features please run:

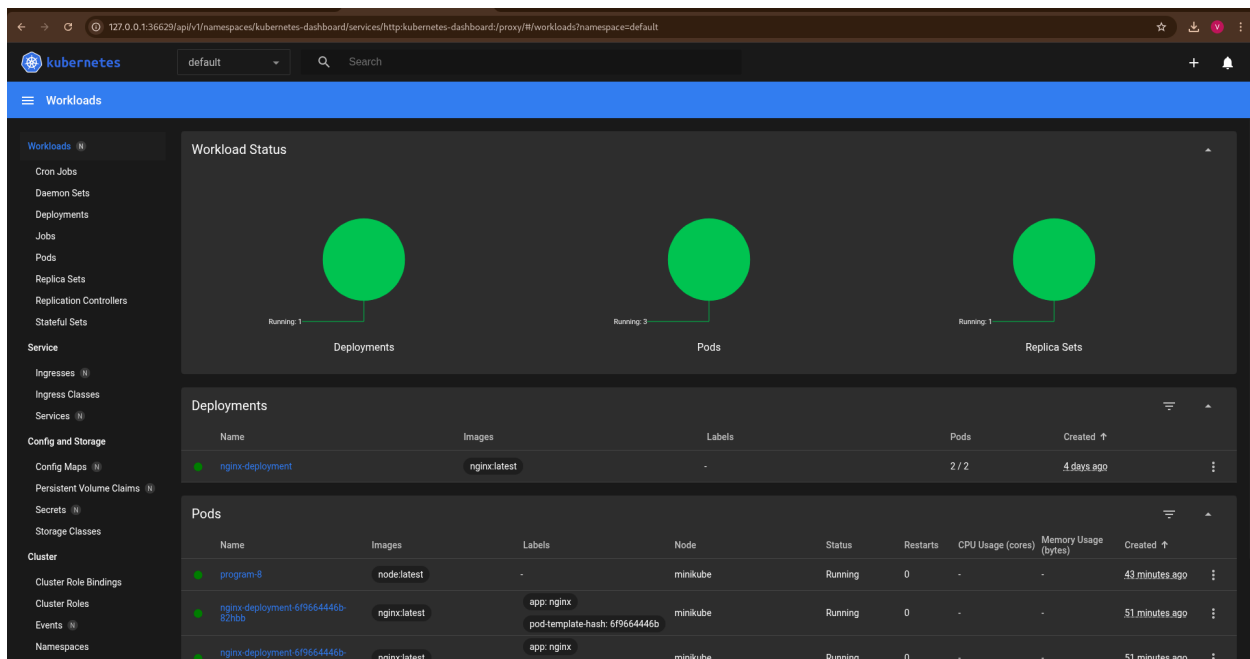
    minikube addons enable metrics-server

🌟 Enabled addons: storage-provisioner, dashboard, default-storageclass
🎉 Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
```

STEP 2 : Open kubernetes dashboard

```
(1RV24MC114_VENKATESHAHD@kali)~[~/mca/devops/lab/lab10]
$ minikube dashboard
👉 Verifying dashboard health ...
🚀 Launching proxy ...
👉 Verifying proxy health ...
🔗 Opening http://127.0.0.1:36629/api/v1/namespaces/kubernetes-dashboard/services/http:kubernetes-dashboard:/proxy/
in your default browser...
Opening in existing browser session.
```

Kubernetes Dashboard



STEP 3 : Verify Running Services

```
(1RV24MC114_VENKATESHAHD@kali)-[~/mca/devops/lab/lab7]
$ minikube status
minikube
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured
```

STEP 4 : CREATE Deployment yamI

```
(1RV24MC114_VENKATESHAHD@kali)-[~/mca/devops/lab/lab7]
$ cat nginx-deployment.yaml
apiVersion: apps/v1
kind : Deployment
metadata:
  name: nginx-deployment

spec:
  replicas: 2
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
      - name: nginx
        image: nginx:latest
        ports:
        - containerPort: 80
```

Step 5 : Apply Deployment

STEP 6 : Check Status

```
(1RV24MC114_VENKATESHAHD@kali)-[~/mca/devops/lab/lab7]
$ kubectl apply -f nginx-deployment.yaml
deployment.apps/nginx-deployment configured
```

```
(1RV24MC114_VENKATESHAHD@kali)-[~/mca/devops/lab/lab7]
$ kubectl get deployments
```

NAME	READY	UP-TO-DATE	AVAILABLE	AGE
nginx-deployment	2/2	2	2	4d1h

```
(1RV24MC114_VENKATESHAHD@kali)-[~/mca/devops/lab/lab7]
$ kubectl get pods
```

NAME	READY	STATUS	RESTARTS	AGE
nginx-deployment-6f9664446b-kc9md	1/1	Running	0	4d1h
nginx-deployment-6f9664446b-zkwjg	1/1	Running	0	4d1h

```
(1RV24MC114_VENKATESHAHD@kali)-[~/mca/devops/lab/lab7]
$ kubectl get replicates
error: the server doesn't have a resource type "replicates"
```

```
(1RV24MC114_VENKATESHAHD@kali)-[~/mca/devops/lab/lab7]
$ kubectl get replicaset
```

NAME	DESIRED	CURRENT	READY	AGE
nginx-deployment-6f9664446b	2	2	2	4d1h

STEP 7 : Access the web app

```
(1RV24MC114_VENKATESHAHD@kali)-[~/mca/devops/lab/lab7]
$ kubectl expose deployment nginx-deployment --type=NodePort --port=80
service/nginx-deployment exposed
```

```
(1RV24MC114_VENKATESHAHD@kali)-[~/mca/devops/lab/lab7]
$ kubectl get services
```

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT(S)	AGE
kubernetes	ClusterIP	10.96.0.1	<none>	443/TCP	18d
nginx-deployment	NodePort	10.109.199.196	<none>	80:30233/TCP	17s
nginx-service	NodePort	10.99.253.172	<none>	80:30813/TCP	4d1h

```
(1RV24MC114_VENKATESHAHD@kali)-[~/mca/devops/lab/lab7]
$ minikube service nginx-deployment
```

NAMESPACE	NAME	TARGET PORT	URL
default	nginx-deployment	80	http://192.168.49.2:30233

Opening service default/nginx-deployment in default browser ...

```
(1RV24MC114_VENKATESHAHD@kali)-[~/mca/devops/lab/lab7]
$ Opening in existing browser session.
```

STEP 8 : DELETE all pods

kubectl delete pods —all